

# Linux

24 November 2024 13:50

## History of Linux:

- Linus Torvalds, a student at the university of Helsinki, Finland Created Linux in 1991.
- Linux is an open source operating System.
- In 1992 the Linux Kernel was Relicensed under the GNU General Public License.
- Linux are command line interface (CLI)
- Linux code written in c language.

## Components of Linux:

There are three main components in Linux

1. Hardware
  2. Kernel
  3. Shell (System utility)
1. Hardware:  
RAM, Hard Disk, CPU they are creating the hardware Layer
2. Kernel:  
Kernel is the most imported part of the Linux operating system. Linux are interact with the hardware Directly. For doing the actual work
3. Shell:  
Shell are to provide the interface between user and kernel (cat /etc/shells)

## Features of Linux:

1. Linux is an open sources operating system.
2. Portability is one of the main features of Linux
3. Linux are multi user and multi programming operating system
4. Multitasking feature is also available in Linux
5. Security is a very important part of any OS this system are very secure vires and malware they are on problem in OS

## Difference between Windows and Linux:

| Windows                                                                             | Linux                                                                                           |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <a href="#">Linux</a> is an open source operating system.                           | While <a href="#">windows</a> are the not the open source operating system.                     |
| Linux is free of cost.                                                              | While it is costly.                                                                             |
| It's file name case-sensitive.                                                      | While it's file name is case-insensitive.                                                       |
| Linux is more efficient in comparison of windows.                                   | While windows are less efficient.                                                               |
| There is forward slash is used for Separating the directories.                      | While there is back slash is used for Separating the directories.                               |
| Linux provides more security than windows                                           | While it provides less security than Linux.                                                     |
| There are 3 types of user account –<br>(1) Regular , (2) Root , (3) Service account | There are 4 types of user account –<br>(1) Administrator , (2) Standard , (3) Child , (4) Guest |

## What is Operating System:

An operating system (OS) is a program that manages a computer's hardware and software resources, while

an application is a program that runs on the OS.

### **What is Application:**

A program that runs on the OS. Applications need an environment to run and perform their tasks. Examples of applications include browsers, MS Office, and games. Application are the bunch of the file

### **What is Desktop OS:**

The desktop OS is the environment in which a user manages a personal computer. It helps in the management of system hardware and software resources. For example, Windows, Mac OS, and various Linux distributions.

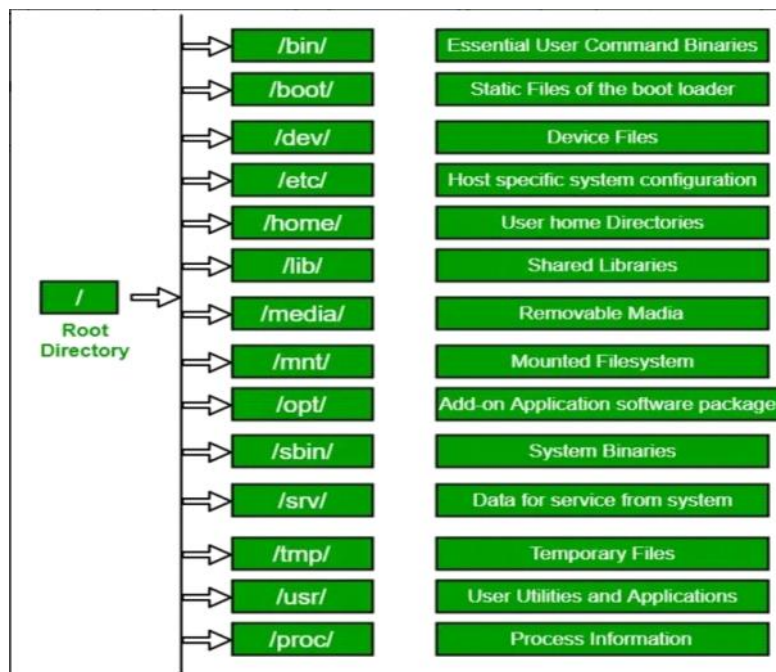
### **What is Server OS:**

It is an operating system designed for usage on servers. It is utilized to give services to a large number of clients. It is a very advanced operating system that can serve several clients simultaneously. Types of server OS: Windows Server, Linux, Unix, MacOS Server

### **What is Client OS:**

It is an operating system that operates within the desktop. It is used to obtain services from a server. It runs on client devices like laptops, and computers and is a very simple operating system. A Client Operating System is designed to run on cease-person devices including desktop computers, laptops, capsules, and smartphones. Types of Client OS: Chrome OS, IOS, Android, BSD(Berkeley Software Distribution)

### **Linux File Hierarchy Structure:**



### **Directory:**

1. /root - home directory of root user.
2. /home - home directory for local user
3. \*\*\* /etc - configuration file (Linux file change kr sakte ho everything are change this is open sources )
4. \*\*\* /bin - (binary) commands file for local user
5. \*\*\* /sbin - (system binary) command file for root user
6. /lib32/64 - (libraries) supporting file
7. /media - USB drivers, printers removable file
8. \*\*\* /mnt - attached, hard disk mount point (empty directory used the disk is using a mount space and disk )
9. /boot - booting file, GRUB, kernel
10. /usr - user related file
11. \*\*\* /var - (variable) mails, logs(apt amazon), web hosting

12. /proc - (process) files related process
13. \*\*\* /temp - (temporary) Files under this directory are deleted when the system is rebooted
14. /ser - (server) server related files
15. /opt - (Optional file) Add-on applications should be installed under either /opt/ or /opt/ sub-directory.
16. /run - running file
17. /dev - devices related file EBC, CPU, Disk, tty, block,
18. /sys - system related file